WORKED SOLUTION

STEP III 2006, Question 14

Probability and Statistics • Independence

STEP III 2006, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q14

The key idea

Everything here runs on two facts: expectation is linear, and for independent variables the expectation of a product is the product of the expectations. Variances come from $\mathrm{Var}(Y) = \mathrm{E}(Y^2) - [\mathrm{E}(Y)]^2$, so we only ever need expectations. To show two variables are *not* independent, it is enough to find one consequence of independence that fails, and the easiest one to test is $\mathrm{E}(UV) = \mathrm{E}(U)\,\mathrm{E}(V)$.

Solution

The two facts

For any random variables and constants a, b,

$$\mathrm{E}(aX_1 + bX_2) = a\,\mathrm{E}(X_1) + b\,\mathrm{E}(X_2),$$

and if X_1 and X_2 are independent then

$$\mathrm{E}(X_1X_2) = \mathrm{E}(X_1)\,\mathrm{E}(X_2).$$

The second fact also applies to functions of the variables: if X_1, X_2 are independent then so are X_1^2 and X_2, X_1 and X_2^2, and X_1^2 and X_2^2. We shall also use $\mathrm{E}(X_i^2) = \sigma_i^2 + \mu_i^2$, which is just $\mathrm{Var}(X_i) = \mathrm{E}(X_i^2) - \mu_i^2$ rearranged.

Perimeter and area

The perimeter is $P = 2X_1 + 2X_2$ and the area is $A = X_1X_2$. By linearity, and by independence,

$$\mathrm{E}(P) = 2(\mu_1 + \mu_2), \qquad \mathrm{E}(A) = \mu_1\mu_2.$$

For the variance of P we work from $\mathrm{E}(P^2)$:

$$\begin{aligned}\mathrm{E}(P^2) &= 4\,\mathrm{E}(X_1^2 + 2X_1X_2 + X_2^2)\\ &= 4(\sigma_1^2 + \mu_1^2 + 2\mu_1\mu_2 + \sigma_2^2 + \mu_2^2)\\ &= 4(\sigma_1^2 + \sigma_2^2) + 4(\mu_1 + \mu_2)^2,\end{aligned}$$

using linearity and $\mathrm{E}(X_1X_2) = \mu_1\mu_2$. Subtracting $[\mathrm{E}(P)]^2 = 4(\mu_1 + \mu_2)^2$,

$$\mathrm{Var}(P) = 4(\sigma_1^2 + \sigma_2^2), \qquad \text{so the standard deviation of } P \text{ is } 2\sqrt{\sigma_1^2 + \sigma_2^2}.$$

For A, since X_1^2 and X_2^2 are independent,

$$\mathrm{E}(A^2) = \mathrm{E}(X_1^2)\,\mathrm{E}(X_2^2) = (\sigma_1^2 + \mu_1^2)(\sigma_2^2 + \mu_2^2),$$

and subtracting $\mu_1^2\mu_2^2$ leaves

$$\mathrm{Var}(A) = \sigma_1^2\sigma_2^2 + \mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2,$$

so the standard deviation of A is

$$\sqrt{\sigma_1^2\sigma_2^2 + \mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2}.$$

P and A are not independent

If P and A were independent we would have $\mathrm{E}(PA) = \mathrm{E}(P)\,\mathrm{E}(A)$. Now $PA = 2X_1^2X_2 + 2X_1X_2^2$, so by linearity and independence of X_1 and X_2,

$$\begin{aligned}\mathrm{E}(PA) &= 2(\sigma_1^2 + \mu_1^2)\mu_2 + 2\mu_1(\sigma_2^2 + \mu_2^2)\\ &= 2(\mu_1 + \mu_2)\mu_1\mu_2 + 2(\mu_2\sigma_1^2 + \mu_1\sigma_2^2)\\ &= \mathrm{E}(P)\,\mathrm{E}(A) + 2(\mu_2\sigma_1^2 + \mu_1\sigma_2^2).\end{aligned}$$

A length and a breadth are positive, so $\mu_1, \mu_2 > 0$, and $\sigma_1, \sigma_2 \neq 0$, so the extra term $2(\mu_2\sigma_1^2 + \mu_1\sigma_2^2)$ is strictly positive. Hence $\mathrm{E}(PA) \neq \mathrm{E}(P)\,\mathrm{E}(A)$, and P and A are not independent.

The variable $Z = P - \alpha A$

Again, independence of Z and A would force $\mathrm{E}(ZA) = \mathrm{E}(Z)\,\mathrm{E}(A)$. By linearity,

$$\begin{aligned}\mathrm{E}(ZA) - \mathrm{E}(Z)\,\mathrm{E}(A) &= [\mathrm{E}(PA) - \mathrm{E}(P)\,\mathrm{E}(A)] - \alpha[\mathrm{E}(A^2) - [\mathrm{E}(A)]^2]\\ &= 2(\mu_2\sigma_1^2 + \mu_1\sigma_2^2) - \alpha(\mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2 + \sigma_1^2\sigma_2^2),\end{aligned}$$

using the two results above. The bracket multiplying α is $\mathrm{Var}(A)$, which is positive (it contains $\sigma_1^2\sigma_2^2 > 0$), so this expression vanishes for exactly one value of α:

$$\alpha_0 = \frac{2(\mu_1\sigma_2^2 + \mu_2\sigma_1^2)}{\mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2 + \sigma_1^2\sigma_2^2}.$$

If $\alpha \neq \alpha_0$, then $\mathrm{E}(ZA) \neq \mathrm{E}(Z)\,\mathrm{E}(A)$, so Z and A are not independent, as required.

Ask yourself Does $\alpha = \alpha_0$ mean that Z and A *are* independent?

No. Zero covariance is a consequence of independence, not a guarantee of it. At $\alpha = \alpha_0$ our test simply gives no information, and we need a different argument. The final part shows that in one concrete case independence fails there too.

The final part

Here $\mu_1 = \mu_2 = 2$ and $\sigma_1^2 = \sigma_2^2 = 1$, so

$$\alpha_0 = \frac{2(2+2)}{4+4+1} = \frac{8}{9}.$$

By the previous part, Z and A are not independent for any $\alpha \neq \frac{8}{9}$. For a single argument that covers every α, list the four equally likely outcomes:

(X_1, X_2)	A	P	$Z = P - \alpha A$
$(1,1)$	1	4	$4 - \alpha$
$(1,3)$, $(3,1)$	3	8	$8 - 3\alpha$
$(3,3)$	9	12	$12 - 9\alpha$

The key observation is that P, and hence Z, is completely determined by A. Let $z = 4 - \alpha$. Knowing $A = 1$ forces $Z = z$, so

$$\mathrm{P}(Z = z,\ A = 1) = \mathrm{P}(A = 1) = \tfrac{1}{4}.$$

Independence would need this to equal $\mathrm{P}(Z = z)\,\mathrm{P}(A = 1)$, that is, $\mathrm{P}(Z = z) = 1$, so Z would have to take the value $4 - \alpha$ on every outcome. That needs $8 - 3\alpha = 4 - \alpha$, so $\alpha = 2$, and also $12 - 9\alpha = 4 - \alpha$, so $\alpha = 1$. These can't both hold, so $\mathrm{P}(Z = z) < 1$ and independence fails.

Hence Z and A are not independent for any value of α. For instance, at $\alpha = \frac{8}{9}$ the values of Z are $\frac{28}{9}$, $\frac{16}{3}$ and 4, and $\mathrm{P}(Z = \frac{28}{9}, A = 1) = \frac{1}{4}$, whereas $\mathrm{P}(Z = \frac{28}{9})\,\mathrm{P}(A = 1) = \frac{1}{16}$.

Checks

- With no variation ($\sigma_1 = \sigma_2 = 0$) both standard deviations reduce to 0, as they should.
- In the final part, $\mathrm{Var}(A) = 1 + 4 + 4 = 9$ by the formula; directly, A takes values $1, 3, 9$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$, giving $\mathrm{E}(A) = 4$, $\mathrm{E}(A^2) = 25$ and variance 9.
- We checked the general mean, variance and covariance formulas exactly, using fractions, on 200 random discrete distributions. We also confirmed that $\mathrm{E}(ZA) = \mathrm{E}(Z)\mathrm{E}(A)$ at α_0, that $\alpha_0 = \frac{8}{9}$ in the final part, and that Z and A fail independence there for $\alpha = \frac{8}{9}$ and 500 other values of α.

Common slips

- Writing $\mathrm{E}(X_1^2) = \mu_1^2$. The variance term σ_1^2 is exactly what makes $\mathrm{Var}(A)$ and the covariance non-zero.
- Concluding that Z and A are independent when $\alpha = \alpha_0$. Zero covariance does not imply independence.
- Forgetting to say why $\mu_2\sigma_1^2 + \mu_1\sigma_2^2 \neq 0$. It is the positivity of lengths that rules out cancellation.

Beyond STEP

The quantity $\mathrm{E}(UV) - \mathrm{E}(U)\mathrm{E}(V)$ is the covariance, and the special value α_0 is $\mathrm{Cov}(P, A)/\mathrm{Var}(A)$, the slope of the least-squares regression of P on A. Choosing $\alpha = \alpha_0$ makes the residual Z uncorrelated with A, but not independent of it. For jointly normal variables the two notions coincide, which is one reason the normal distribution is so convenient. Grimmett and Welsh, *Probability: An Introduction*, treats covariance, correlation and independence carefully.

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